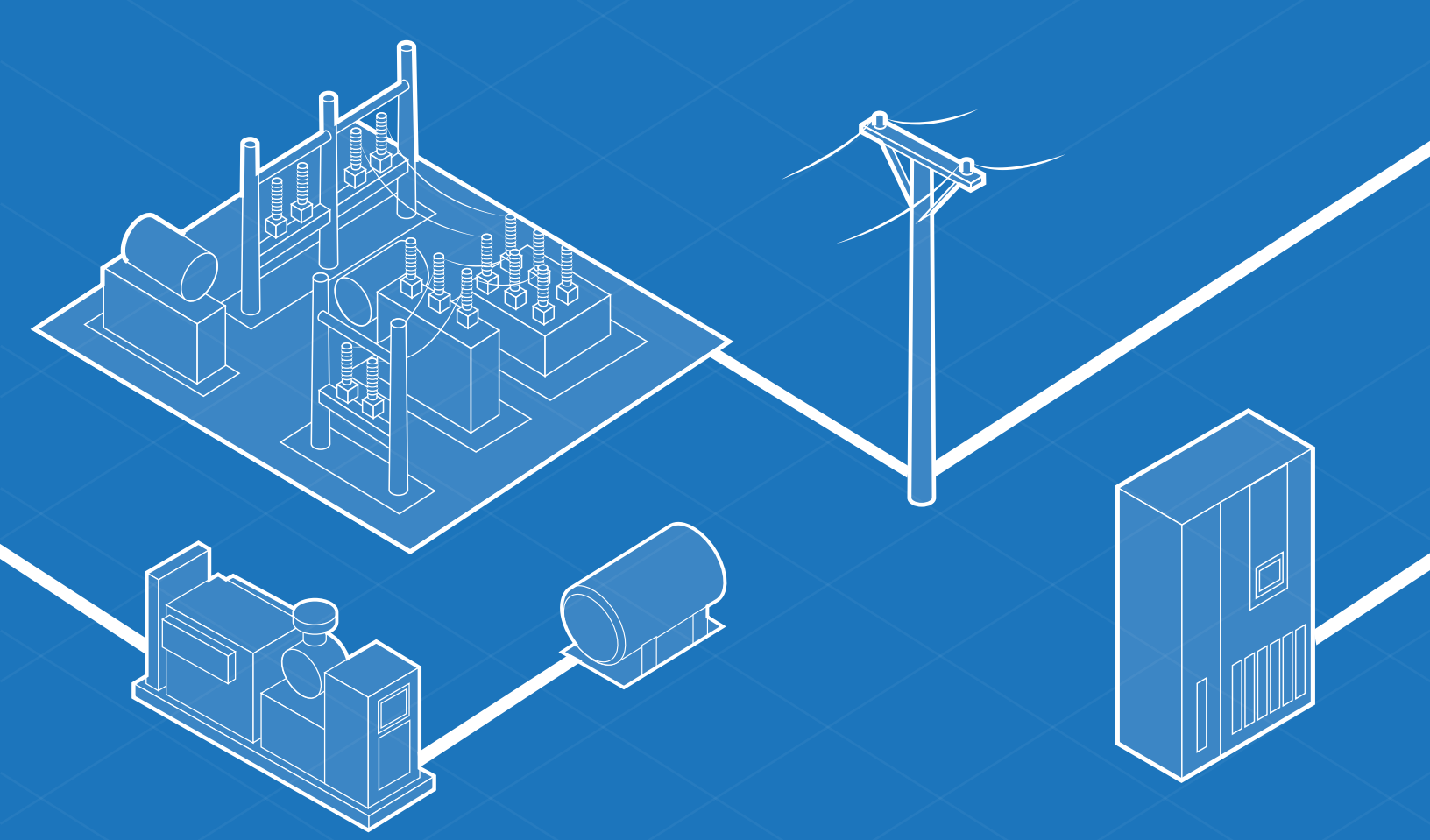


DATA CENTER

POWER COMPARTMENTALIZATION

For business organizations that rely heavily or even exclusively on their online presence and/or the availability of their data to conduct business, the cost of downtime has become essentially incalculable – a prolonged or poorly timed outage could potentially be fatal to the business. Many organizations are choosing to put their availability and data protection needs into the hands of experts at data centers. A select few of these data center choose to comply with TIA-942-A 2012 standards for design and assurance of reliability.

Plenty of US data centers make ambiguous claims about their "Tier 4" infrastructure but rarely specify:

TIA-942-A 2012

If their claims are measured against TIA-942-A 2012 or other industry standards

If their facilities have been audited by a knowledgeable third party

Whether their assertions encompass all four infrastructure elements (Telecommunications, Architectural, Electrical, and Mechanical)

A data center can have different Tier ratings for different elements of infrastructure, e.g. "Tier 4 Electrical and Mechanical, Tier 3 Architectural." A data center's overall rating is only as high as its lowest rated element, so the overall data center rating in the previous example would be a Tier 3. The "lowest = overall" rating model makes ambiguous ratings claims that much more misleading and meaningless.

One of the most important concepts in Tier 4 power delivery systems and in ultra-high availability in general is "compartmentalization." In this discussion, we will explain the concept of data center power compartmentalization and why it matters.

WHAT IS POWER COMPARTMENTALIZATION?

As defined by Lifeline Data Centers: "Data center power compartmentalization is the practice of isolating or segregating complementary power systems and paths from one another to ensure that each disparate part of the infrastructure is independent from all others and performs its designated tasks on its own."

The word "Compartmentalization" does not appear anywhere in the TIA-942-A 2012 standard. However, strict adherence to the prescriptive requirements for Tier 4 power delivery in the TIA-942-A 2012 standard will result in a fully compartmentalized design outcome.

BENEFITS:

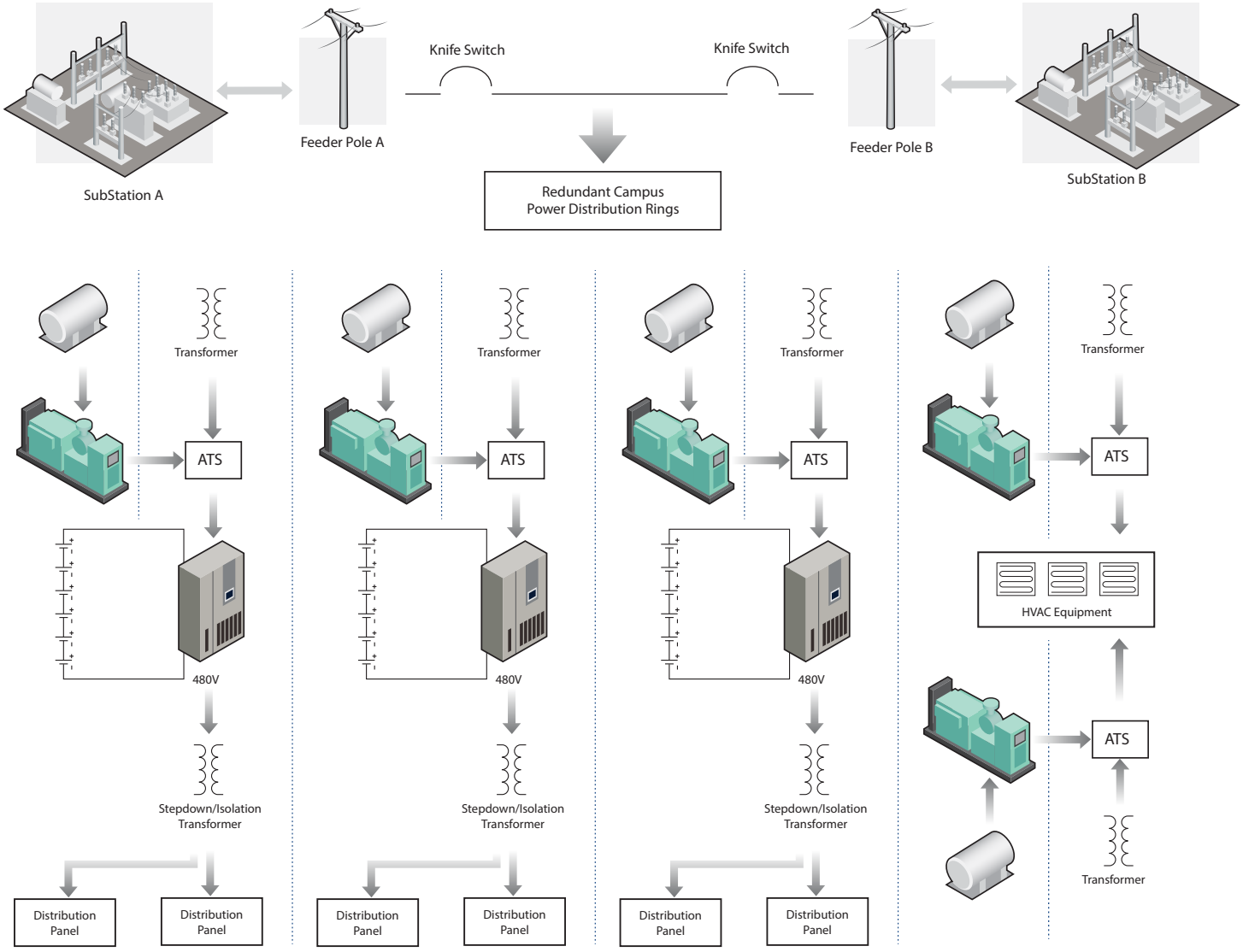
- Simplicity of design - A fully compartmentalized power delivery system is often called a "Simple Current Path" or "Simple Power Path" delivery system.
- Brute force uptime – A failure in one power delivery system or path will not affect any other power delivery path.

DRAWBACKS:

- Expensive to build and maintain – the infrastructure is by definition 100% redundant; EVERYTHING is duplicated. This is a capital-intensive endeavor.
- Inherent power inefficiencies – there is twice as much infrastructure as required to service the electrical load at every level of the delivery system. The result is idle capacity, which is not optimally efficient.

THE RIGHT WAY TO DO POWER COMPARTMENTALIZATION

Effective power compartmentalization has a basic rule of thumb: no commonality between feeds or sharing of infrastructure. In the setup illustrated below, two disparate MAJOR failures need to occur in order to totally disrupt power to an end-user's equipment, assuming the end-user is taking full advantage of redundant power feeds via dual power supplies in their equipment, etc. An effective integration of fully compartmentalized Tier 4 power delivery infrastructure and proper deployment and configuration of customer equipment WILL realize the Tier 4 promise of 99.995% uptime, or better.



Some highlights of this specific configuration:

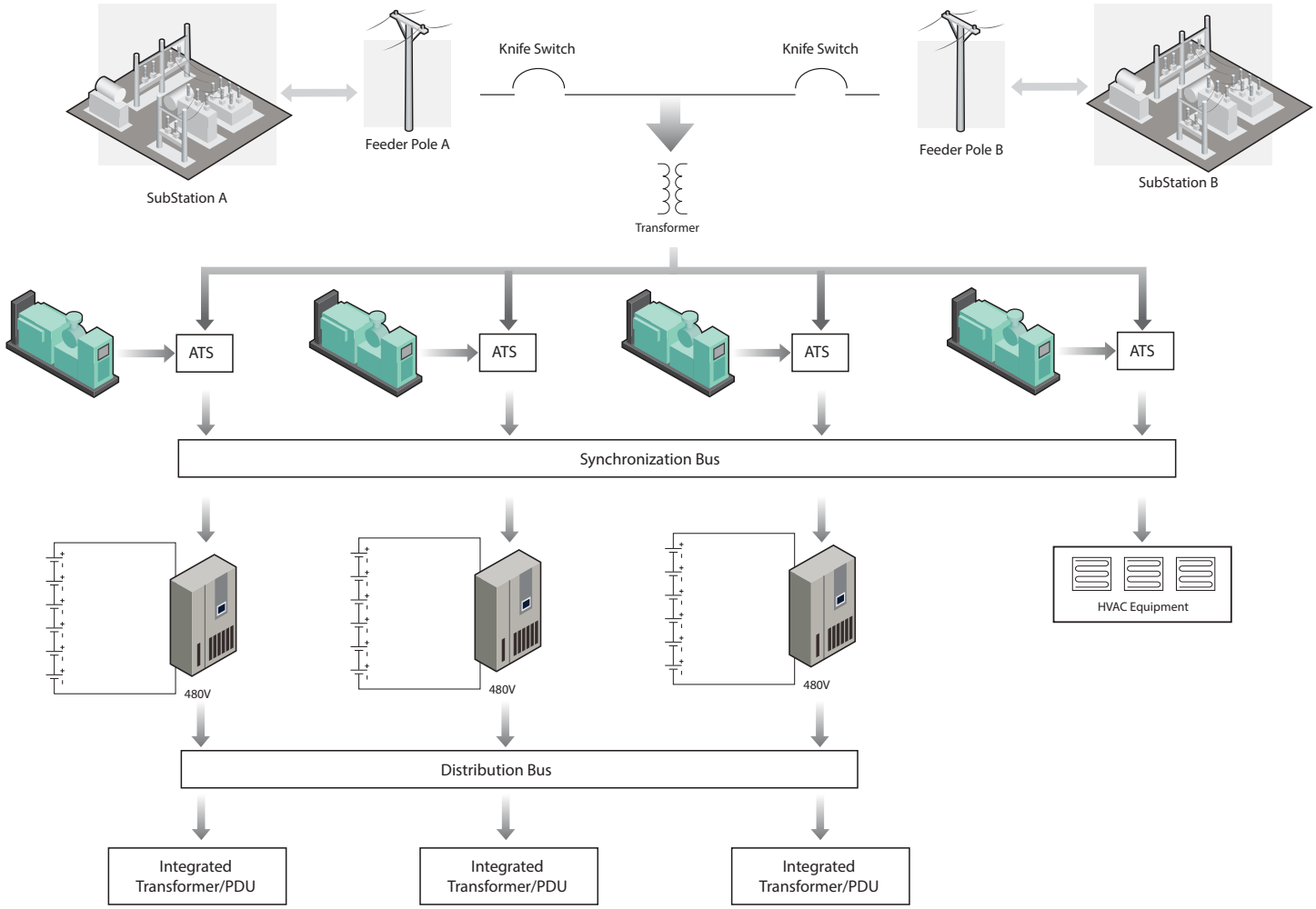
- The ability to attach to either of two utility substations
- Dedicated redundant generator systems for HVAC equipment to prevent mixing HVAC equipment loads and ICT (Information & Communications Technology) equipment loads on the same generators.
- 480V 3ϕ power runs are extended as far as possible into the data center's ICT envelope to minimize the length of single ϕ runs. 3ϕ power runs cancel their own electromagnetic noise; single ϕ power runs generate electromagnetic noise that is exceptionally difficult to shield or filter.

Highlights of compartmentalization as depicted in this illustration:

- The customer's redundant power feeds are fed power by:
  - Separate utility company transformers.
  - Generator equipment housed in separate buildings and supplied by separate fuel delivery systems.
  - Completely disparate UPS/Battery systems – there is absolutely no commonality or interaction between the delivery systems.
- On the output side of the UPS systems the step-down/isolation transformers and distribution panels are physically separate components, not traditional integrated transformer/breaker panel PDU systems.
- The UPS systems are free to accept power from their dedicated generators as soon as the power meets that UPS system's criteria – there is no synchronization delay.

WHAT HAPPENS WITHOUT POWER COMPARTMENTALIZATION

There are fundamental problems associated with having a non-compartmentalized power delivery system in the data center, primarily centered on commonality of components in delivery paths and outright single points of failure.



A note about the drawing above: The synchronization bus and distribution bus architecture illustrated in this drawing is frequently used in large data centers such as those that Amazon and others build and operate. Sadly, many smaller data centers don't even rise to the level of this architecture – far too many are haphazard designs that have evolved over time, or, "we've always done it that way", or have nothing to do with any published data center design standards at all.

Some lowlights of this configuration:

- The delivery systems are fed from the same utility company transformer. One blown transformer fuse or catastrophic transformer failure will force all backup power delivery systems into service.
- The UPS systems cannot accept power from the generator systems until the synchronization bus presents the power to the UPS systems. This results in a much longer period of time on battery than the simple current path design. The more deeply a battery is discharged, the more likely it is to fail catastrophically, and the shorter the battery's operating life.
- One balky generator or generator control system can prevent power from being presented to the UPS systems at all. (See Amazon story below)
- The synchronization bus is a complex piece of hardware and a single point of failure.
- The generator systems for HVAC equipment are servicing mixed HVAC equipment (inductive) loads and ICT equipment (resistive) loads.
- The presence of the HVAC equipment inductive load on the synchronization bus subjects the inputs of the UPS systems to the inherently unpredictable voltage variations and harmonic distortions associated with inductive electrical loads upon sudden startup.

AN EXAMPLE OF WHY COMPARTMENTALIZATION MATTERS

In June 2012, Amazon.com experienced an outage due to reliance on a synchronization bus design, resulting in hours of downtime. For a business as big as Amazon, one hour of downtime can lead to millions of dollars lost in sales and, possibly, customers. An excerpt from Amazon's statement about their June 6, 2012 outage:

"At 7:24pm PDT, a large voltage spike was experienced by the electrical switching equipment in two of the US East-1 datacenters supporting a single Availability Zone. All utility electrical switches in both datacenters initiated transfer to generator power. In one of the datacenters, the transfer completed without incident. In the other, the generators started successfully, but each generator independently failed to provide stable voltage as they were brought into service. As a result, the generators did not pick up the load and servers operated without interruption during this period on the Uninterruptable Power Supply ("UPS") units. Shortly thereafter, utility power was restored and our datacenter personnel transferred the datacenter back to utility power. The utility power in the Region failed a second time at 7:57pm PDT. Again, all rooms of this one facility failed to successfully transfer to generator power while all of our other datacenters in the Region continued to operate without customer impact.

In the single datacenter that did not successfully transfer to the generator backup, all servers continued to operate normally on Uninterruptable Power Supply ("UPS") power. As onsite personnel worked to stabilize the primary and backup power generators, the UPS systems were depleting and servers began losing power at 8:04pm PDT."

For the full story on the Amazon outage, visit http://bit.ly/amazonout.

WHAT CAN YOU DO?

Ask your prospective or current data center provider to discuss power compartmentalization and how they've incorporated it into their power delivery infrastructure. If they look at you as though you're from Uranus or some other planet, you're probably not talking to the right data center, IaaS, or Cloud Services provider.

If you're looking for a data center designed and operated by knowledgeable and certified data center professionals who understand and use power compartmentalization effectively, schedule a tour of Lifeline Data Centers today:

http://bit.ly/tourldc